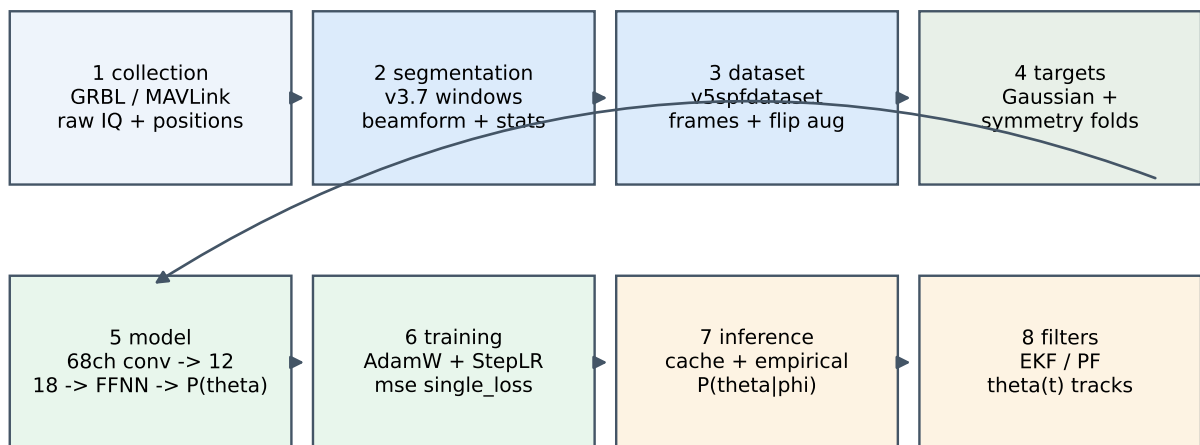


The SPF Processing Pipeline

A Reviewed, Verified End-to-End Summary

collection · segmentation · dataset · targets · model · training · inference · filters

The SPF processing pipeline (stages reviewed in this report)



Every claim in this report traces to the adversarially verified documentation in `claude_docs/` (inventories, deep dives, KNOWN_ISSUES, and the stage-1 audit), spot-checked against source at commit 574c454.

Generated from `tutorials/build_pipeline_report.py`

1 Executive summary

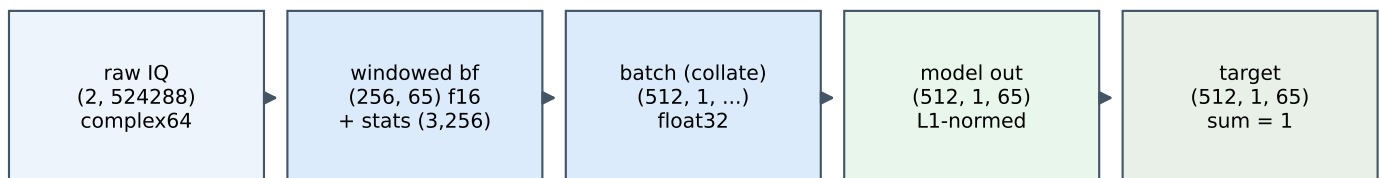
SPF localizes RF emitters by measuring the inter-antenna phase difference $\phi = 2\pi(d/\lambda)\sin\theta$ at two 2-element receivers and carrying that phase through eight stages: physical capture, windowed beamforming, a dataset layer with symmetry augmentation, a small neural model that outputs a 65-bin distribution over angle, and time-fusion filters.

VERDICT: the pipeline is internally consistent and correct on its active paths. The stage-1 training path was audited end-to-end (all shape contracts verified; reflections proven correct; both scheduler/key hazards inapplicable) and the latest models reproduce their predecessors exactly (single val 0.0984 vs 0.0984). The main risks are NOT math errors -- they are silently-wrong side paths (a cache-key collision, a segmentation filter that drops abutting signal), rover-safety gaps, and thin direct test coverage on the most load-bearing numerics.

Top cross-stage findings

- The production model NEVER sees the segmentation mask: it consumes the raw 256-window beamformer. So the headline segmentation bug (#45, dropped abutting signal) skips the NN path -- but lands on `mean_phase`, exactly what the empirical $P(\theta|\phi)$ table and all non-NN filters key on.
- One sign convention is load-bearing across three stages: the dataset's $-\sin$ target, `get_phase_diff` returning $-\phi$, and every filter negating antenna spacing must agree; none of the three is pinned by a test.
- d/λ spans 0.122--1.549 in production: a large fraction of the fleet operates ABOVE the 0.5 unambiguous limit, so ϕ aliases multiple θ . The learned 65-bin distribution + device/spacing embeddings quietly do ambiguity resolution the analytic $\arcsin$ cannot -- likely part of the +36% BladeRF OOD gap.

Shape flow for the stage-1 training path (batch 256 x 2 radios = 512 rows)



2 Key numbers (verified)

carrier f_c (`rx_lo`) → 868/915 MHz, 2.412--2.467 GHz,
5.766--5.866 GHz

antenna spacing d → 0.025--0.080 m (rover prod 50.75 mm)

d/λ across datasets → 0.122 – 1.549 (> 0.5: aliased regime)

sample rate / RF bandwidth → 30 MHz / 3 MHz (prod)

buffer per snapshot → 524,288 samples (2^{19}) × 2 elements, complex64
 window size / stride / windows → 2048 / 2048 / 256 per snapshot
 angle grid n_θ → 65 bins on $[-\pi, \pi]$
 segmentation version → 3.7 (trim 20%, stddev ≤ 0.5 , min seg 3000, min |sig| 40)
 batch → 256 sessions → 512 collated rows (×2 radios)
 optimizer → AdamW; single lr 1e-4 wd 1e-3; paired lr 2e-4 wd 0
 scheduler → StepLR γ 0.5, step 6 (single) / 12 (paired), per epoch
 target → Gaussian σ 0.1 rad, $\pm\pi$ wrap-fold + front/back fold
 model size → conv 64ch → 12 + FFNN 18 → 512 depth 4 (thin)
 jun26 single (final) → val/single 0.0984 @ 1.875M steps (== dec15)
 jun26 paired (running) → val/paired 0.1517 @ 1.05M vs dec15 0.1615 @ 2.75M
 BladeRF OOD gap → +36% (0.1323 vs 0.0971 mean per-dataset val loss)
 uniform-prediction floor → val/uniform ≈ 0.129 (context for 0.0984)

3 Stage 1 -- Data collection (raw IQ -> .zarr)

Two platform families move radios through space while recording raw IQ + ground-truth positions: the WALL ARRAY (CNC gantry, GRBL steppers, XY in mm, v5 format) and the ROVER (ArduPilot over MAVLink, GPS, v4 format). Both drive a motion routine (bounce/circle/center/diamond) and record two 2-element receivers plus an emitter (ESP32 WiFi / LoRa / VTX / SDR-TX). Per-receiver threads read the SDR, tag position+time, and write the zarr; the AD9361 RX1-RX2 phase-inversion register calibration is load-bearing for coherence.

output → signal_matrix (snapshots, 2, 524288) c64 + per-snapshot metadata
 v5 adds → tx/rx positions (mm) + rx_heading_in_pis; v4 instead
 GPS + heading (degrees)
 collection size → 600,000 records per receiver (rover prod)

Issues (verified): #42 GRBL out-of-bounds exception kills the motion thread while collection keeps stamping the FROZEN position (silently wrong data); rover safety cluster #40 (EKF arm gate omits HORIZ_ABS), #43 (unmapped mode KeyError strands planner), #44 (RTL no timeout), #18 (sudo shutdown from handler thread). Real serial/MAVLink hot paths are untested (fake-hardware tests only).

4 Stage 2 -- Segmentation (v3.7: .zarr -> .yarr)

Decides which 2048-sample windows contain signal vs noise and precomputes everything training needs, so the NN never touches raw IQ. Per session: detrend → per-window delay-and-sum beamform against $e^{-j2\pi(d/\lambda)\sin\theta}$ steering → trimmed circular mean/stddev of ϕ + median |signal| → threshold/merge/drop → boolean mask + weighted session stats.

outputs (per receiver, f16 cache) → windowed_beamformer (snap, 256, 65) · all_windows_stats (snap, 3, 256)

plus (f32) → weighted beamformer/stats, bool mask, mean_phase (NaN if no signal)

correctness note → several properties silently require stride == window_size (true in prod)

Issues: #45 keep_signal_surrounded_by_noise drops BOTH of two abutting signal runs the merge declined to join -- fires on the default config; hits mean_phase (empirical table + filters), not the NN input. #37 the live ceil-rank percentile is untested while the tested variant has no callers. Coverage: the windowed-beamformer writer and the live circular-stats engine have NO direct tests (audit P0 gaps).

5 Stage 3 -- Dataset layer (v5spfdataset)

Reads zarr + yarr + yaml, computes ground-truth angles from positions, and serves torch tensors. Three angle frames: y_rad = array-relative, craft_y_rad = craft-heading-relative, absolute_theta = world. The ϕ target y_phi carries the $-\sin$ sign convention that matches get_phase_diff. Augmentation in render_session: FLIP (50%: negate θ / ϕ , flip the beamformer θ -axis + stats ch0 -- verified consistent across every key); DOUBLE_FLIP (paired stage: $\theta \rightarrow \text{sign}(\theta)\pi - \theta$).

collate → paired rows stacked flat: B = 256 × 2 = 512; f16 upcast to f32

stage-1 batch keys → windowed_beamformer, all_windows_stats, gains, spacing, freq, vehicle, sdr, targets

quirk → empirical is force-appended despite empirical_input false (shape/device only)

Issues: #7 two live breakpoint()s on the only realtime/absolute-north path (P0); #32/#39/#33 realtime-dataset races (unlocked read vs eviction, init race, thread leak); #11 mutable-default mutation. Coverage: ground-truth angles tested (zero-heading only); the FLIP augmentation has NO test (audit P0 gap).

6 Stage 4 -- Targets and symmetry

Ground-truth θ becomes a 65-bin distribution: an analytic Gaussian in radians ($\sigma = 0.1$), built on a 3-block extended grid so mass crossing $\pm\pi$ folds to the correct bins, L1-normalized. For the single model the target is averaged with its front/back reflection $\theta \rightarrow \text{sign}(\theta)\pi - \theta$ -- valid because $\sin(\text{sign}(\theta)\pi - \theta) = \sin \theta$ gives the identical ϕ , so the supervision is deliberately bimodal.

Symmetry composition (proven in the audit): left/right is handled as INPUT augmentation (flip), front/back in the TARGET (the π -fold); different reflections at different stages, so they compose with no double-fold. scatter_k : 21 in the configs is INERT on this path (kernel blur belongs to the unused onehot path).

Coverage: NONE direct -- no assertion pins unit-sum, the wrap-fold, or the fold symmetry (audit P0 gap; proposed tests exist in the plan).

7 Stage 5 -- Model

Stage 1, SinglePointWithBeamformer: AllWindowsStatsNet concatenates the 3 stats channels + 65 beamformer channels → (B, 68, 256), an 8-layer Conv1d(68 → 64) stack
→ 12 window features (with train-only window augmentation: fraction 0.5, dropout 0.25, shuffle 0.15, shrink 0.5). PrepareInput appends 6 scalars (gains 2, spacing, freq, vehicle, sdr) → 18-dim → FFNN depth-4 hidden-512 (layer norm)
→ abs → L1-norm over the angle axis → single: (B, 1, 65).
Both hard shape contracts (68-channel, 18-dim) were verified against source.

Stage 2, PairedSinglePointWithBeamformer: runs the (detached) single net on both radios, rotates radio-0's distribution into the craft frame, concatenates, and a fusion FFNN outputs the craft-relative paired distribution; initialized from stage-1 best.pth (load_single).

inputs NOT used → raw IQ, phase stats, session-averaged beamformer, empirical (all flags false)

scale caveat → beamformer enters raw (no normalization) scaled by a hardcoded /500

Issues: #34 PrepareInput mutates vehicle_type batch views in place every training step (benign single-consumer, fragile). Coverage: the CI training fixture uses a DIFFERENT model variant (phase input, no windows_stats_net) -- the production forward path is never exercised by tests (the audit's keystone gap).

8 Stage 6 -- Training (and the latest runs)

Exactly ONE term is backpropagated per stage: mse_loss(single, target) (stage 1) or the paired equivalent (stage 2) -- SSE over 65 bins between two unit-sum distributions. Everything else logged (craft/aligned/rand/uniform losses) is no_grad diagnostics. fp32 throughout, AMP off. best.pth is selected on val loss; resume parses checkpoint step names.

jun26 single → val/single 0.0984 @ 1.875M -- exact tie with dec15 (0.0984): faithful reproduction

jun26 paired → val/paired 0.1517 @ 1.05M, still improving -- ahead of dec15's best-ever

dec15 paired precedent → overfit after ~epoch 12 (0.1548 → 0.1615); watch for the turn

OOD → BladeRF val +36% above non-blade -- excluded from training (noblade split)

Issues: #1 cosine-scheduler branch crashes (UnboundLocalError) but only one abandoned config selects it; numpy RNG is NOT seeded (torch/random only) so seed 10 does not fully pin the data path. Coverage: loss fn, optimizer, and the production train path have no direct tests.

9 Stage 7 -- Inference and the empirical baseline

Checkpoints run over datasets into an .npz cache keyed dataset-basename / seg-version / checkpoint-md5 / config-md5; the NN particle filters consume this cache. The non-learned baseline is the empirical $P(\theta|\phi)$ table: a symmetrized 2-D histogram of (true θ , observed mean_{phase}) per device+spacing group, column-normalized; the producer-consumer join was verified end-to-end.

Issues: #41 the cache key OMTS the v4 flag (which changes outputs) and keys on basename only -- colliding runs silently return wrong cached inference (P1, silently-wrong). #47 never-invalidated .md5 sidecars compound it for in-place checkpoint edits. #48 the empirical builder is broken for EVEN bin counts -- and the CLI default is 50; production artifacts dodge it with 65/7. Coverage: cache-key correctness and the empirical builder have no tests.

10 Stage 8 -- Filters (EKF / PF)

Fuse per-snapshot angle evidence into smooth $\theta(t)$ tracks. Observation model $\phi = 2\pi(d/\lambda)\sin(\theta - \text{offset})$ per radio; every concrete filter NEGATES the spacing at construction to match the dataset's $-\sin$ convention (load-bearing, untested). Motion: constant velocity. Likelihoods: empirical $P(\theta|\phi)$ lookup, or the cached NN distribution indexed by each particle's θ -bin.

active classes → EKF single/dual θ , PF single/dual θ , PF single-NN (5 of 8)

broken / excluded → PF dual-NN (live breakpoint, #2) · XY variants commented out of prod grid

Issues: #2 the wired-in dual-radio-NN PF hits an unconditional breakpoint() on step 0 -- under the prod multiprocessing pool the worker hangs forever (P0, one-line fix). #3 the whole B2/DynamoDB cloud harness is un-importable since a module move (~14 months, 2-line fix). #31 return_particles=True crashes (.copy() on a torch tensor). Coverage: single/dual paths tested on fake data with loose bounds; edge paths untested.

11 Conventions that hold the pipeline together

- Angles: radians on $[-\pi, \pi]$, normalized by pi_norm (plain modulo mishandles negatives). ϕ = observable phase difference; θ = angle of arrival; $\phi = 2\pi(d/\lambda)\sin \theta$ with the global $-\sin$ / negated-spacing sign pact.
- Orientations stored as multiples of π (*_in_pis); v4 rover heading is DEGREES by design (the v4 → v5 converter bridges it -- do not 'fix', it is load-bearing).
- Positions: zarr stores mm; dataset tensors emit meters. rx_spacing in meters; rx_wavelength_spacing = d/λ dimensionless.
- Vocabulary: snapshot (one capture, one receiver) $\supset$ 256 windows of 2048 samples; session = 1 snapshot in prod; 'paired' = both radios of one craft.
- A 2-element array cannot separate θ from $\pi - \theta$; the two receivers are mounted 90° apart precisely so pairing resolves it.

12 ~~Review: top issues and test-coverage state~~

Top issues by severity x reachability (all adversarially verified)

1. #2 P0 active -- live breakpoint() in the wired-in dual-radio-NN PF; prod pool worker hangs on step 0.
2. #7 P0 active -- two breakpoint()s on the only realtime/absolute-north inference path (headless rover hang).
3. #41 P1 silent -- inference cache key omits v4 + basename-only keying: wrong cached outputs on collision.
4. #45 P1 silent -- segmentation drops abutting signal runs on the default config (hits mean_phase/filters).
5. #42 P1 silent -- GRBL out-of-bounds kills motion thread; collection stamps frozen positions thereafter.
6. #40 P1 safety -- rover EKF arm gate duplicates HORIZ_REL, omits HORIZ_ABS.
7. #43/#44 P1 safety -- unmapped mode KeyError strands planner; RTL/move has no effective timeout.
8. #3 P0 dormant -- entire B2 cloud filter pipeline un-importable (module move); 2-line fix.
9. #32/#39 P1 -- realtime dataset races (KeyError under prod settings; init AttributeError).
10. #47 P2 -- stale .md5 sidecars silently reuse old inference caches (compounds #41).

Test-coverage verdict (from the stage-1 audit matrix)

All 11 stage-1 segments are CORRECT as written, but the most load-bearing pieces have the weakest direct coverage: the flip augmentation, the target folds, the live circular-stats engine, and the windowed-beamformer writer have NO direct tests, and the CI training fixture exercises a non-production model variant. A zero-prod-change test plan exists (claude_docs/reference/_stage1_audit/TEST_PLAN.md): keystone prod-shaped smoke test first, then flip-consistency, target-fold symmetry, and circular-stats tests.

Where to go deeper: claude_docs/README.md is the index -- per-stage overviews (01--05), verified per-function contracts (reference/spf/*), the canonical issue list (KNOWN_ISSUES.md), the stage-1 audit (reference/_stage1_audit/), and the companion physics tutorial (tutorials/spf_signal_path.pdf).